

Technology Stack

Date	28 June 2026
Team ID	xxxxxx
Project Name	Rising Waters
Maximum Marks	2 Marks

Define Your Technology Stack:

The **Raising Waters** flood prediction system leverages a **modern full-stack architecture** with **React.js** for an interactive frontend dashboard, **Python with FastAPI** for high-performance backend APIs, and **PostgreSQL** for structured data storage. The machine learning pipeline utilizes **TensorFlow** and **Scikit-learn** for training predictive models (LSTM, Random Forest, XGBoost) on weather data, with **Redis** enabling real-time caching and **Celery** for asynchronous task processing. The system is deployed on **AWS cloud infrastructure** using **Docker containers**, with **Twilio** for SMS alerts, **Leaflet.js** for flood risk mapping, and **Prometheus/Grafana** for comprehensive system monitoring and performance tracking.

Technology Stack Details

S.No	Architecture Component / Layer	Technology Chosen	Justification / Purpose
1	Frontend / Client-Side	React.js, Leaflet.js, D3.js, Tailwind CSS	React.js provides a dynamic and responsive user interface for real-time flood dashboards. Leaflet.js enables interactive flood risk mapping with geospatial visualization. D3.js creates powerful data visualizations and trend charts. Tailwind CSS ensures rapid UI development with consistent, modern styling across all devices.

2	Backend / Server-Side	<i>Python (FastAPI), Node.js</i>	FastAPI offers high-performance asynchronous API handling with automatic OpenAPI documentation, making it ideal for real-time weather data processing. Node.js complements for handling WebSocket connections and real-time notifications. Both provide excellent scalability and speed for concurrent user requests.
3	Database / Data Storage	<i>PostgreSQL, TimescaleDB, Redis</i>	PostgreSQL ensures reliable ACID compliance for user management and structured data. TimescaleDB (PostgreSQL extension) efficiently handles time-series weather data with native time-based partitioning. Redis provides ultra-fast caching for frequently accessed prediction results and real-time data, reducing database load.
4	Cloud / Hosting / Deployment	Amazon Web Services (AWS): EC2, S3, RDS, Lambda	AWS was chosen for its robust scalability, global availability, and extensive service ecosystem. EC2 provides flexible compute capacity for ML model inference. S3 offers cost-effective storage for historical weather data and model artifacts. RDS simplifies managed PostgreSQL database operations. Lambda enables serverless processing for lightweight data ingestion tasks, reducing operational costs.

5	Version Control & CI/CD	GitHub, GitHub Actions	<i>GitHub enables seamless team collaboration with branch protection and code review workflows. GitHub Actions automates the CI/CD pipeline with continuous testing, building, and deployment to AWS, ensuring rapid and reliable delivery of updates with minimal human error.</i>
6	Third-Party APIs / Other Tools	OpenWeatherMap API, IMD API, Twilio, SendGrid, Docker/	OpenWeatherMap API provides real-time and historical weather data for flood prediction. IMD API (India Meteorological Department) ensures locally relevant and authoritative weather information. Twilio enables instant SMS alerts to emergency teams and public users. SendGrid handles automated email notifications and reports. Docker containerizes the application for consistent deployment across development and production environments.